

The S^6 record

a reader of its results

Verification and reconstruction of a claimed complex structure on S^6
(Zenodo record 10.5281/zenodo.22678442)

An audit and reading prepared by Claude (Anthropic),
model `claude-opus-5-5` (Opus 5.5), at maximum reasoning effort

Record: KokunoYumeto, with the AI contributors credited below; the construction is that of the
manuscript circulated by Levent Alpöge

27 September 2026 (version 2, after one independent referee pass)

Companion to the reader of the split-zero zeta programme, version 2, DOI 10.5281/zenodo.22977243

Abstract

The Zenodo record 10.5281/zenodo.22678442, *Verification and Reconstruction of a Claimed Complex Structure on S^6 : An AI-Produced Validate/Repair/Disprove Record*, collects the owner's AI-produced audit of a manuscript circulated by Levent Alpöge. The manuscript claims that a family of complex two-dimensional tori over the projective line, completed at three special fibres, is a complex structure on the six-sphere. The record holds 33 files from four versions (1–9 September 2026): a 154-page verification monograph, the reconstructed and annotated manuscript, an eleven-page correction note on a step in a published theorem of Campana, Demailly and Peternell, topology supplements, and a 174 MB archive of the full project.

This reader maps the record, states what it claims, with its own labels, and says how far each claim was checked here. The central claims are:

- the construction gives a compact complex threefold diffeomorphic to S^6 , of algebraic dimension one;
- it is therefore a counterexample to the published theorem;
- a specific step of that theorem's proof is not a formal consequence of its displayed sequences. The source manuscript had already made this point for its own fibre; the record adds an independent test family.

The first two are claims of the record and of the source manuscript. An independent exposition by Philip Engel (13 September 2026) presents the construction with the stated aim of a self-contained proof. Neither the construction nor the counterexample claim was checked here.

What this reader does check (Section 5):

- the correction note, in full. Its test family shows that the step in question needs an extra hypothesis. Added here:
 - the step is valid on normal fibres, as the source manuscript already notes, and the torsion that breaks it is present at every non-normal point;
 - at a reduced fibre with normal crossings, the group that the step asserts to vanish is the space of sections of the line bundle pulled back to the normalization. So the step fails exactly when that bundle has a nonzero section;
 - in the test family this group is one-dimensional for every gluing parameter;
- the local algebra behind one of the record's corrections to the source manuscript, in general form. The correction is right for relative one-forms, but relative two-forms do have torsion at the cusp fibre;
- a number of finite statements, and the exact set of values of the cubic in the key advance S^6 -5.

This version includes the findings of one independent referee pass (Appendix B).

About this reader

The record, and who made it.

- The candidate construction and its original claims are attributed by the record to their reported source. Different files name the source differently:
 - the monograph’s title page says “Claude/Fabel”;
 - the key-advances reader of 6 September says the construction was circulated by Levent Alpöge with Fable;
 - the later attribution file of the workbench (ATTRIBUTION.md of the Yang–Mills repository, 21 September), which its author designates as the clarification, says the manuscript was circulated by Levent Alpöge and produced with Claude.
- The verification monograph describes itself as Codex’s mathematical verification, with two independent ChatGPT Pro review records. The record calls itself an AI-produced verification aid, not specialist certification, and makes no priority claim (record README.md).
- The record’s creator, in its metadata, is KokunoYumeto. Its provenance file records that the user of the underlying Codex session directed that the programme be called validate/repair/disprove, and that it put proofs, calculations and counterexamples before verdicts.
- The correction note’s author line reads “human author to be supplied”.

What was read.

- The record’s files, downloaded on 26 September 2026 with the owner’s permission. Their MD5 checksums equal the record’s; the 174 MB archive 28_ was not downloaded.
- The correction note, in full.
- From the monograph:
 - its orientation (Part I), including its 21-item status list;
 - its Theorem 13.6;
 - its closing dispositions (§§13.11 and 14).

Its Theorems 13.5, 13.25 and 13.28 are cited by the numbers the record gives them.

- From the manuscript: its opening summary, its Remarks 1.3 and 9.21, and its §§10.1–10.3 (the quotation of the published step, Lemma 10.3, Remark 10.4, and the statement and first steps of Theorem 10.5).
- The claim ledger in the record’s evidence archive 04_.
- The five-page key-advances reader, the reader’s guide to the 5 September supplement, the guide to the frozen project, and the provenance file.

The rest was mapped from its tables of contents and summaries.

How it was checked, and the labels. The labels are those of the companion readers, with one addition:

- Proved here: the proof is given in full in this reader;
- Audited: the record’s proof was read in full, every step was checked, and independent checks were run;
- Reproduced: the finite content was recomputed;
- Located: the statement and its proof were found and are described; nothing was checked;
- Stated: the statement and the location of its proof are reported, but the proof was not seen.

The record's own labels are VERIFIED, OPEN and REFUTED, assigned by its AI verification. They are quoted as the record's and are not endorsed here. This version includes the findings of one independent referee pass, each verified before it was applied (Appendix B).

Contents

About this reader	1
1 The question, the published obstruction, and what the record claims	5
1.1 The question	5
1.2 The claimed construction, and its reception so far	5
1.3 What the record says about its own status	5
2 The record, and what each file contains	6
2.1 Four versions under one concept DOI	6
2.2 Where to start	6
2.3 What is in the conversation files	6
3 The claimed construction, in outline	6
4 The record's ledger and its central claims	7
4.1 The central claims (record: VERIFIED)	8
4.2 The four rows labelled REFUTED	8
4.3 The two rows labelled OPEN	9
4.4 Confirmed local repairs	9
5 The correction note on the singular-fibre step, audited	9
5.1 The filtration and the exact criterion	10
5.2 Where the torsion lives	10
5.3 The test family, audited and sharpened	13
5.4 What survives: the derived global criterion	15
5.5 Relative differentials on the local models	15
6 The later supplements, 5–9 September	17
6.1 The topology supplement of 5 September (version 1.2)	17
6.2 The key-advances reader of 6 September	18
7 Where the record meets the owner's other workbenches	20
8 What remains open, and what was not checked	21
8.1 Open questions	21
8.2 What this reader did not check	21
A Catalogue of statements and files	22
A.1 Statements	22
A.2 Files	23

B	How this reader was checked	24
	References	26

1 The question, the published obstruction, and what the record claims

1.1 The question

Whether the six-sphere carries a complex structure (an integrable almost complex structure) has long been open. Campana, Demailly and Peternell studied a hypothetical complex structure X on S^6 through its algebraic dimension $a(X)$, the transcendence degree of its field of meromorphic functions, first in 1998 [4] and then in 2020 [5]. In the 2020 paper they prove that a compact complex threefold homeomorphic to S^6 would have algebraic dimension 0 ([5], Corollary 2.3; Engel’s exposition cites the same statement from the 1998 paper, [4], Corollary 4.2). A complex structure on S^6 that fibres holomorphically over $\mathbb{P}^1$ has $a(X) \geq 1$, and would therefore contradict that theorem.

1.2 The claimed construction, and its reception so far

- *The source manuscript* [1], whose opening summary is titled *The $(3, 4, \infty)$ modular family of 2-tori, completed at its three special points, is a complex structure on S^6* , claims exactly such a threefold: $X \rightarrow \mathbb{P}^1$ with $a(X) = 1$, X diffeomorphic to S^6 . It is unrefereed; the record reconstructs it as file 03_ (108 pages). The workbench’s attribution file says that it was circulated by Levent Alpöge and produced with Claude.
- *This record* (24 August to 9 September 2026) is an AI-produced audit of that manuscript. It reconstructs the manuscript, repairs local defects, and concludes, with its own label VERIFIED, that the constructed threefold satisfies every hypothesis of the relevant theorem of [5] and contradicts its conclusion. It also isolates a step in the published proof that is not a formal consequence of the sequences displayed there. The source manuscript had already located the failure of that step at its own cusp fibre; the record’s correction note adds an independent test family (Sections 4 and 5).
- *An exposition.* Philip Engel’s *Complex structures on S^6* (13 September 2026) [6] presents the construction, which it credits to Claude under Alpöge’s direction, with the stated aim of a self-contained proof. It notes that the construction contradicts the result that the algebraic dimension must be 0, which it cites as [4], Corollary 4.2, and it describes the construction as a one-parameter family of threefolds diffeomorphic to S^6 .

This reader does not assess the global construction or the counterexample claim, and takes no position on them. It reports what the record claims and checks what can be checked from the record’s own finite and local statements.

1.3 What the record says about its own status

- It is an AI-produced verification aid. It makes no priority claim, and its labels do not constitute journal certification or community acceptance (record README.md).
- At the freeze of version 1.0 its internal claim ledger (in the evidence archive 04_) had 42 rows labelled VERIFIED, none CONDITIONAL, 2 OPEN and 4 REFUTED. The two open rows are:
 - independent specialist replication of the cyclic invariant lattices, the logarithmic determinant divisor, the residue cancellation and an order-four scalar test;
 - specialist and community adjudication of the complete construction and counterexample.

The README describes them as replication and external adjudication, not missing internal premises.

- The later supplements (5–9 September) state that they do not independently establish a complex structure on S^6 or disprove the published theorem. They use the construction’s data as stated input.

These statements are accurate descriptions of the documents read.

2 The record, and what each file contains

2.1 Four versions under one concept DOI

The record 10.5281/zenodo.22678442 is the latest of four versions of the concept record 10.5281/zenodo.22235523. Each version kept the files of the one before and added new ones (read from the record's public metadata on 26 September 2026):

Version (DOI 10.5281/zenodo....)	Files added	What was added
22235524, 1 Sept. (1.0)	20	The 154-page verification monograph (01_); the 128-page annotated manuscript (02_); the 108-page clean reconstruction of the source manuscript (03_); sources and evidence (04_, 20.8 MB); a mathematical provenance file and its receipt (05_, 06_); a sanitised conversation record and a typed workflow annex, with receipts and validation reports (07_–13_); README, licence and rights files
22238779, 1 Sept. (1.1)	2	The eleven-page correction note on the singular-fibre step of [5], with its source (00_)
22346462, 5 Sept. (1.2)	6	A topology supplement: a reader's guide (14_), <i>The higher torus wrap on the cusp</i> (15_, 50 pp.), <i>Real S^{12} and S^{24} routes</i> (16_, 64 pp.), their sources and notes (17_–19_)
22678442, 9 Sept. (consolidated)	5	The five-page key-advances reader of 6 September and its source (26_, 27_); the complete frozen project of 6 September (28_, 174 MB, 1,079 files, including a 782-page cumulative workbench and a 138-page higher-rung paper); a guide and a manifest (29_, 30_)

The record's front file README.md was written for version 1.0 and describes it. The later guides are 14_s6_topology_update_readers_guide_2026-09-05.pdf and 29_S6_FROZEN_PROJECT_GUIDE_2026-09-09.md.

2.2 Where to start

- *For the claimed construction:* the clean reconstruction 03_ (108 pages). Its opening pages summarise the whole argument.
- *For the audit:* the monograph 01_. Its Part I (five pages) orients; its Part II is the complete validate/repair/disprove paper, with its ledger and its dispositions (§§13.11 and 14).
- *For the published theorem:* the correction note 00_ (eleven pages), which is self-contained. Section 5 states and audits it.
- *For the later work:* the key-advances reader 26_, and the guide 14_ to the topology supplement.

2.3 What is in the conversation files

Files 07_–13_ hold a sanitised record of the human–assistant conversation (976 events) and a typed workflow annex (20,977 events), with receipts and validation reports. The record's README describes their redaction and validation and says that the private raw session record is not deposited. This reader did not use these files.

3 The claimed construction, in outline

This section follows the opening summary of the reconstructed source manuscript (file 03_) and the monograph's orientation. Nothing in it was checked here, apart from the arithmetic noted.

1. *Base.* The triangle group $\Delta = \Delta(3, 4, \infty) \cong (\mathbb{Z}/3) * (\mathbb{Z}/4)$ acts on the upper half-plane. The quotient, with its cusp added, is $\mathbb{P}^1$ with orbifold points of orders 3 and 4 and one cusp.
2. *A lattice representation.* On $V \cong \mathbb{Z}^4$ there are automorphisms T_1 of order 3 and T_2 of order 4 whose product has unipotent inverse T_0 with $(T_0 - I)^2 = 0$; Δ acts on the dual lattice Λ through them.
3. *Periods and the family.* Period functions τ, μ, β on the half-plane give an equivariant period map. The monograph writes the period matrix as

$$\Pi = [Z \mid I_2], \quad Z = \begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}, \quad D = \operatorname{Im} \beta - \frac{6(\operatorname{Im} \mu)^2}{\operatorname{Im} \tau} < 0, \quad \det_{\mathbb{R}} \Pi = \operatorname{Im} \tau \cdot D < 0.$$

The quotient is a family of complex two-dimensional tori over the punctured orbifold curve. Its very general fibre is not algebraic, and its Néron–Severi group is generated by one indefinite class.

4. *Three fillings.* At the two elliptic points, Kodaira’s logarithmic transform: $(\text{disc} \times \text{torus})/(\mathbb{Z}/m)$, $m = 3, 4$, giving multiple fibres. At the cusp, Mumford’s toric degeneration, whose central fibre is a degree-six del Pezzo surface glued to itself along opposite sides, with two triple points.
5. *Gluing and the fundamental group.* The fillings are glued along collars, and the gluings can be changed by translations. The fundamental group and the homology depend on three integers ℓ_0, ℓ_1, ℓ_2 , with

$$\pi_1(X) \cong \mathbb{Z}/|12\ell_0 - 4\ell_1 - 3\ell_2|.$$

The record’s candidate takes $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, for which $12\ell_0 - 4\ell_1 - 3\ell_2 = -1$, so the fundamental group is trivial (source manuscript §7; monograph Part I).

6. *Conclusions claimed.* For this choice, the manuscript concludes:

- X is simply connected with the integral homology of S^6 ;
- X is therefore diffeomorphic to S^6 , as there are no exotic smooth structures on the six-sphere;
- $c_3(X) = 2$, localised at the cusp;
- $a(X) = 1$, with the fibration as algebraic reduction.

The period data of item 3 are also what the owner’s Yang–Mills and Erdős–Straus workbenches take from this construction (Section 7). In the Yang–Mills record the quantity $D_{\text{YM}} = L_T q_T + 6m_T^2$, with $L_T = \operatorname{Im} \tau$, $q_T = -\operatorname{Im} \beta$ and $m_T = \operatorname{Im} \mu$, equals $-\operatorname{Im} \tau \cdot D = -\det_{\mathbb{R}} \Pi > 0$. So the two records use the same number, with opposite sign conventions.

4 The record’s ledger and its central claims

The record labels each claim with its own status: VERIFIED for a complete proof, OPEN when neither a proof nor a counterexample is recorded, and REFUTED for an exact assertion with a recorded contradiction or counterexample. These are the AI verification’s labels. Two lists carry them, and this reader cites both:

- the *claim ledger* in the evidence archive 04_ with 48 rows: 42 VERIFIED, 2 OPEN, 4 REFUTED. The README quotes its counts.
- the monograph’s *status list*, the 21 numbered items of its Part I. Item numbers below refer to this list.

4.1 The central claims (record: VERIFIED)

- *The construction.* According to the monograph, its sections 6–7 prove the period-family descent, the toric and finite fillings, the global gluing, and the integral nearby-cycle and Leray comparisons. With them the constructed threefold X satisfies $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = 0$, $a(X) = 1$ and $c_3(X) = 2$, and $f : X \rightarrow \mathbb{P}^1$ is holomorphic and an algebraic reduction.
- *A counterexample to the published theorem.* X satisfies every stated hypothesis of Theorem 2.2 of [5] and contradicts its conclusion (monograph §13.11, item 5). The monograph says that this uses the complete construction and topology proofs, not only the two defects below.
- *A corrected criterion.* This is the part of the published argument that survives. If some line bundle L_0 has $H^2(X, T_X \otimes L_0) = 0$, then generic vanishing holds and $c_3(X) \leq 0$ (monograph Theorem 13.28; the correction note’s Theorem 5.2). For the constructed X this hypothesis fails for every L_0 , since $c_3(X) = 2$.
- *Further invariants:* the Hodge diamond and the Frölicher sequence, the Bott–Chern and Aeppli dimensions, and global representatives with the Bott–Chern product structure (status list, items 17, 19 and 20).

Status. Located. Not checked here. They are the substance of the record, and they need specialist verification; the record itself lists that adjudication as open.

4.2 The four rows labelled REFUTED

Each row is given with its claim-ledger identifier.

1. *The singular-fibre inference of [5]* (p. 692 of the journal version; S6-CDP-P692; status list, item 7).
 - The inference: for an irreducible reduced fibre, the published proof reduces a vanishing to the vanishing of $H^0(S, \widetilde{\Omega}_S^1 \otimes L|_S)$.
 - The record says that this is not a formal consequence of the two cotangent sequences, and gives a test family.
 - The source manuscript had already located the failure at its own cusp fibre (its Lemma 10.3 and Theorem 10.5), and had noted that the step is valid on a normal fibre (its Remark 10.4).

Status. The test family and the corrected criterion are audited here. The exact size of the failure at fibres with normal crossings, and the exact condition for the torsion that causes it, are proved here (Section 5). The source manuscript quotes the published sentence verbatim (its §10.1). The paper itself was not re-read here.

2. *A homology formula imported from [4]* (S6-CDP98-MON0; status list, item 12). The direct-product formula of the 1998 paper (its Lemma 3.2, in the record’s reading) omits the coinvariants of the monodromy. This leaves a torus branch of the 2020 argument without its printed rank argument.

Status. Located.

3. *A remark of the source manuscript on relative differentials* (S6-HODGE-CUSP-TORSION; status list, item 18).
 - The manuscript’s Remark 1.3 speaks of the sheaves $\Omega_{X/B}^p$, and its Remark 9.21 of $\Omega_{X/B}^1$, as having torsion along all three singular fibres.
 - The record shows that at the cusp $\Omega_{X/B}^1$ is torsion-free but not locally free, and that its torsion occurs only at the two multiple fibres.

Status. Proved here for the local models the record gives (Proposition 5.10). The refutation is correct for $p = 1$. For $p = 2$ the same local models do have torsion along the double curves of the cusp fibre, so Remark 1.3's statement holds for $\Omega_{X/B}^2$.

4. *A remark of the source manuscript on the gluing integers* (S6-LOCAL-728; monograph §14, item 2). The remark (its Remark 7.28) is stated for a fixed triple. The triple $(\ell_0, \ell_1, \ell_2) = (100, 1, -1)$ gives $12\ell_0 - 4\ell_1 - 3\ell_2 = 1199 \neq \pm 1$, so the statement must quantify existentially over ℓ_0 for fixed admissible (ℓ_1, ℓ_2) .

Status. Reproduced: the arithmetic.

4.3 The two rows labelled OPEN

- S6-OPEN-HODGE-REPLICATION: independent specialist replication of the cyclic invariant lattices, the logarithmic determinant divisor, the residue cancellation and an order-four scalar test. The monograph's status list has this as its single open item (item 21), together with the three cotangent direct images and the values $h^{1,2} = 1, h^{1,1} = 2$.
- S6-OPEN-EXTERNAL-ADJUDICATION: specialist and community adjudication of the complete construction and counterexample.

The monograph states that these rows are open because independent human replication and adjudication are outstanding, and that they are not unresolved premises of its proofs.

4.4 Confirmed local repairs

Monograph §14 lists seven items, labelled separately from the theorem. Five are labelled VERIFIED:

- a closed-image argument;
- an appendix sequence whose arrows are cohomological although it is labelled as homology;
- a fundamental-group identification;
- a build defect of the record's own assembler;
- a correction to a finite-filling computation of a canonical section.

The other two are the REFUTED remarks treated above. The five are located.

5 The correction note on the singular-fibre step, audited

The correction note (file 00_, eleven pages, dated 1 September 2026, added in version 1.1) isolates one step in the singular-fibre part of the proof in [5] and records what survives without it. Its scope statement is narrow: it does not infer that Theorem 2.2 of [5] is false, and it credits the source manuscript for the conductor-supported differential on which it builds (the manuscript's §10.2, Lemma 10.3). The source manuscript had already located the failure of this step at its own cusp fibre (its remark on [5] after the main theorem, and its Theorem 10.5). In its Remark 10.4 it had also recorded that the step is valid when the fibre is normal.

This section states the note's results and reports the audit. It adds four things, each proved here:

- the exact condition for the torsion that makes the step fail: it is present exactly at the non-normal points (Proposition 5.3);
- a description of the kernel sheaf on which the step depends. At a reduced fibre with normal crossings, the group that the step asserts to vanish is the space of sections of the line bundle pulled back to the normalization (Proposition 5.5). This was found in the referee pass of this reader.

- the exact size of the failure in the note's test family (Propositions 5.7 and 5.5);
- the general form of the note's local computation of relative differentials, including second exterior powers (Proposition 5.10).

Status. Audited. Every proof in the note was read in full and every step checked, and the finite and symbolic content was recomputed (`s6r_checks.py`, Appendix B). No mathematical error was found.

5.1 The filtration and the exact criterion

Let X be a complex manifold and $S \subset X$ a reduced effective Cartier divisor, with smooth locus $j : S_{\text{reg}} \hookrightarrow S$ and conormal sheaf $N^* = N_{S/X}^*$. Put

$$T_S := \ker(\Omega_S^1 \rightarrow j_* \Omega_{S_{\text{reg}}}^1), \quad \widetilde{\Omega}_S^1 := \Omega_S^1 / T_S, \quad K_S := \ker(\Omega_X^1|_S \rightarrow \widetilde{\Omega}_S^1).$$

By the note's Lemma 1.1, T_S is the torsion subsheaf of Ω_S^1 .

Proposition 5.1 (the note's Lemma 2.1 and Propositions 2.2–2.3).

- (a) The conormal sequence $0 \rightarrow N^* \rightarrow \Omega_X^1|_S \rightarrow \Omega_S^1 \rightarrow 0$ is exact on the left.
- (b) There are exact sequences $0 \rightarrow N^* \rightarrow K_S \rightarrow T_S \rightarrow 0$ and $0 \rightarrow K_S \rightarrow \Omega_X^1|_S \rightarrow \widetilde{\Omega}_S^1 \rightarrow 0$.
- (c) Let A be a vector bundle on S with $H^0(S, \widetilde{\Omega}_S^1 \otimes A) = 0$ and $H^0(S, N^* \otimes A) = 0$. Then

$$H^0(S, \Omega_X^1|_S \otimes A) \cong \ker(\delta_A : H^0(S, T_S \otimes A) \rightarrow H^1(S, N^* \otimes A)),$$

where δ_A is the connecting map of the first sequence in (b) tensored with A .

Proof. (a) Locally $S = \{g = 0\}$ with g reduced in a regular local ring, and the map is $\bar{h} \mapsto h dg$. The differential dg is nonzero at the general point of every component of S , since each component is generically smooth and g generates its ideal there. So $h dg \in g \Omega^1$ forces h to vanish on every component, and $h \in (g)$ because (g) is radical. (b) K_S is the preimage of T_S under the surjection $\Omega_X^1|_S \rightarrow \Omega_S^1$, whose kernel is N^* by (a). (c) Tensor both sequences of (b) with A and take cohomology. The second gives $H^0(\Omega_X^1|_S \otimes A) \cong H^0(K_S \otimes A)$, and the first gives $H^0(K_S \otimes A) \cong \ker \delta_A$. $\square$

Status. Audited. The note states (c) for a line bundle on a threefold; the proof uses neither restriction. Reducedness is needed in (a): for $g = g_0^2$, the class of $g_0 \cdot g$ in N^* is nonzero, and it maps to $g_0 dg = 2g dg_0$, which is zero in $\Omega_X^1|_S$.

Thus the implication at issue,

$$H^0(S, N^* \otimes A) = 0 \text{ and } H^0(S, \widetilde{\Omega}_S^1 \otimes A) = 0 \implies H^0(S, \Omega_X^1|_S \otimes A) = 0,$$

holds whenever $T_S = 0$, since then $K_S = N^*$. In general it holds exactly when δ_A is injective (under its two hypotheses).

5.2 Where the torsion lives

Lemma 5.2 (modules with one relation). *Let R be a Noetherian ring with total ring of fractions Q , and let $v \in R^n$ be a vector whose entries generate an ideal J containing a nonzerodivisor. Then $\varphi \mapsto [\varphi v]$ induces an isomorphism*

$$(R :_Q J) / R \xrightarrow{\sim} \text{Tors}(R^n / Rv), \quad \text{and} \quad (R :_Q J) / R \cong \text{Ext}_R^1(R/J, R).$$

In particular, when $J \neq R$, the module R^n / Rv is torsion-free if and only if J has grade at least 2; when $J = R$ it is free.

Proof. For $\varphi \in (R :_Q J)$ the vector φv lies in R^n . If f is a nonzerodivisor with $f\varphi \in R$, then $f[\varphi v] = [(f\varphi)v] = 0$, so the class is torsion. It is zero iff $\varphi v = cv$ for some $c \in R$, that is, iff $(\varphi - c)J = 0$ in Q ; since J contains a unit of Q , this means $\varphi = c \in R$. Conversely, if $f[a] = 0$ for a nonzerodivisor f , then $fa = hv$ with $h \in R$; then $\varphi = h/f$ satisfies $\varphi v = a \in R^n$, so $\varphi \in (R :_Q J)$ and $[a]$ is its image.

For the second isomorphism, apply $\text{Hom}_R(-, R)$ to $0 \rightarrow J \rightarrow R \rightarrow R/J \rightarrow 0$. Here $\text{Hom}(R/J, R) = 0$, since J contains a nonzerodivisor, and $\text{Ext}^1(R, R) = 0$. Every homomorphism $\psi : J \rightarrow R$ is multiplication by $\psi(f_0)/f_0 \in Q$ for any nonzerodivisor $f_0 \in J$. So $\text{Hom}(J, R) = (R :_Q J)$, and the sequence gives $(R :_Q J)/R \cong \text{Ext}^1(R/J, R)$. By Rees's theorem [10, Thm. 16.6], the grade of J is the least i with $\text{Ext}^i(R/J, R) \neq 0$. The grade is at least 1, and it is at least 2 iff Ext^1 vanishes. $\square$

Proposition 5.3 (torsion of the differentials of a hypersurface). *Let $S = \{g = 0\}$ be a reduced hypersurface in a complex manifold of dimension $n \geq 2$. Let $x \in S$, let $R = \mathcal{O}_{S,x}$, and let $J \subset R$ be the ideal generated by the restrictions of $\partial g / \partial z_1, \dots, \partial g / \partial z_n$. Then*

$$T_{S,x} \cong \text{Ext}_R^1(R/J, R),$$

and $T_{S,x} = 0$ iff $\text{ht } J \geq 2$. Equivalently, $T_{S,x} = 0$ iff the singular locus of S has codimension at least 2 in S at x , that is, iff S is normal at x . When a component of $\text{Sing } S$ through x has codimension one, $T_{S,x} \neq 0$. Globally, let $\mathcal{J} \subset \mathcal{O}_S$ be the Jacobian ideal sheaf, generated locally by the restrictions of the $\partial g / \partial z_i$. Then $\varphi \otimes [g] \mapsto [\varphi dg]$ is a canonical isomorphism

$$\text{Ext}_{\mathcal{O}_S}^1(\mathcal{O}_S/\mathcal{J}, \mathcal{O}_S) \otimes N_{S/X}^* \xrightarrow{\sim} T_S.$$

Proof. The differentials have the presentation $\Omega_{S,x}^1 = R^n/Rv$ with $v = (\partial g / \partial z_i|_S)_i$.

J contains a nonzerodivisor. The ring R is reduced, so its zerodivisors form the union of its minimal primes. No minimal prime contains J , because $dg \neq 0$ at the general point of each component. Prime avoidance then gives an element of J outside all minimal primes.

The torsion. Lemma 5.2 gives $T_{S,x} \cong \text{Ext}^1(R/J, R)$, which vanishes iff $\text{grade } J \geq 2$.

Grade, height and codimension. R is a regular local ring modulo a nonzerodivisor, hence Cohen–Macaulay. So $\text{grade } J = \text{ht } J = \dim R - \dim R/J$ [10, Thm. 17.4]. A point $y \in S$ is smooth iff $dg(y) \neq 0$: if S is smooth at y , its ideal is generated by one coordinate function w , so g is a unit times w . Hence the zero set of J is $\text{Sing } S$, and $\dim R/J$ is the dimension of the germ of $\text{Sing } S$ at x [7]. Since R is Cohen–Macaulay, it satisfies Serre's condition S_2 . So, by Serre's criterion [10, Thm. 23.8], it is normal iff it is regular in codimension one, that is, iff the singular locus has codimension at least 2 (for normal complex spaces see also [7]).

The global form. Replace g by ug with u a unit. On S , $\partial(ug)/\partial z_i = u \partial g / \partial z_i$, so $\mathcal{J}$ does not change. The form φdg , for $\varphi \in (\mathcal{O}_S :_Q \mathcal{J})$, is holomorphic and does not depend on coordinates, and $(\varphi/u)d(ug) = \varphi dg$ on S . So the local isomorphisms of Lemma 5.2, $\varphi \mapsto [\varphi dg]$, glue once the generator $[g]$ of $N_{S/X}^*$ is carried along. The identification $(R :_Q J)/R \cong \text{Ext}_R^1(R/J, R)$ of Lemma 5.2 is canonical. $\square$

Corollary 5.4. *The implication at issue holds on every normal reduced fibre (for a surface fibre: one whose singular points are isolated). On a non-normal reduced fibre, $T_S \neq 0$, and the implication holds exactly when δ_A is injective.*

The first statement is the source manuscript's Remark 10.4, proved there by the second Riemann extension theorem. Proposition 5.3 adds the converse: on a reduced hypersurface the torsion of Ω_S^1 is present at exactly those points where the singular locus has a component of codimension one. The fibre of the candidate threefold to which the step would be applied is the cusp fibre, which the record describes as a degree-six del Pezzo surface glued to itself along curves. The

corollary therefore does not cover it, as the record says. On any reduced fibre $N_{S/X}^* \cong O_S$ (on a multiple fibre it is a nontrivial torsion bundle), so the map on which the step depends is

$$\delta_A : H^0(S, \mathcal{E}xt_{O_S}^1(O_S/\mathcal{J}, O_S) \otimes A) \rightarrow H^1(S, A),$$

a map between groups defined by the fibre and A alone. For a fibre with normal crossings, such as the candidate's cusp fibre, $\mathcal{J}$ is the reduced ideal of the double curve.

For $p = 1$, Proposition 5.3 is a case of theorems of Vetter [18] and Lebelt [13], summarized in [15]. Let a reduced complete intersection have singular locus of codimension k . Then Ω^p is torsion-free for $p < k$ and has torsion for $p = k$ (Vetter), and conversely torsion-freeness forces $p < k$ (Lebelt). The proof above is elementary and gives the torsion as an Ext sheaf.

Proposition 5.5 (the kernel sheaf). *Let $S = \{g = 0\}$ be a reduced hypersurface in a complex manifold X , with Jacobian ideal sheaf $\mathcal{J}$.*

- (a) *The map $\varphi \otimes [g] \mapsto \varphi dg$ is an isomorphism $\mathcal{H}om_{O_S}(\mathcal{J}, O_S) \otimes N^* \rightarrow K_S$, which restricts to $O_S \otimes N^* \cong N^*$. So the first sequence of Proposition 5.1(b) is the sequence $0 \rightarrow O_S \rightarrow \mathcal{H}om(\mathcal{J}, O_S) \rightarrow \mathcal{E}xt^1(O_S/\mathcal{J}, O_S) \rightarrow 0$, tensored with N^* .*
- (b) *Suppose S has normal crossings: locally g is a unit times x , xy or xyz in suitable coordinates. Let $\eta : \tilde{S} \rightarrow S$ be the normalization. Then $\mathcal{H}om(\mathcal{J}, O_S) = \eta_* O_{\tilde{S}}$. Hence, if $H^0(S, \tilde{\Omega}_S^1 \otimes A) = 0$ for a line bundle A on S , then*

$$H^0(S, \Omega_{X|S}^1 \otimes A) \cong H^0(\tilde{S}, \eta^*(N^* \otimes A)).$$

- (c) *On a reduced fibre with normal crossings and with $H^0(S, \tilde{\Omega}_S^1 \otimes A) = 0$, the conclusion $H^0(S, \Omega_{X|S}^1 \otimes A) = 0$ of the step holds if and only if $H^0(\tilde{S}, \eta^* A) = 0$.*
- (d) *In the note's test family, $H^0(S, \Omega_{X|S}^1 \otimes A) \cong \mathbb{C}$ for every $\lambda \in \mathbb{C}^*$. For $\lambda = 1$ it is spanned by dt .*

Proof. (a) Work at x with R and v as in Proposition 5.3, and identify $\Omega_{X|S}^1$ at x with R^n through the dz_i .

- A vector a lies in $K_{S,x}$ iff $[a]$ is torsion.
- By Lemma 5.2 this holds iff $[a] = [\varphi v]$ for some $\varphi \in (R :_Q J)$. Since $R \subseteq (R :_Q J)$, this holds iff $a \in (R :_Q J) v$.
- The map $\varphi \mapsto \varphi v$ is injective, since J contains a unit of Q .

So $K_{S,x} = (R :_Q J) dg \cong (R :_Q J) = \text{Hom}_R(J, R)$, and $R dg$ is N^* . The choices do not matter, as in Proposition 5.3.

(b) Choose coordinates in which $g = x_1 \cdots x_k$ with $k \leq 3$, the unit being absorbed into x_1 . The branches are $S_i = \{x_i = 0\}$, and $\tilde{R} = \prod_i \mathbb{C}\{\hat{x}_i\}$ is the ring of tuples of functions on the branches.

- *R inside $\tilde{R}$.* R consists of the tuples that agree on the pairwise intersections $S_i \cap S_l$. For $k = 2$ this is the note's Lemma 3.1, for $k = 3$ the manuscript's Lemma 10.2.
- *The generators of J .* The generator $\prod_{j \neq i} x_j$ of J vanishes on every branch but S_i . So, for $\varphi = (\varphi_l) \in Q$, the product $\varphi \prod_{j \neq i} x_j$ lies in R iff $\varphi_i \prod_{j \neq i} x_j$ is holomorphic on S_i and vanishes on each $S_i \cap S_l$.
- *The quotient.* The x_j with $j \neq i$ are distinct primes of $\mathbb{C}\{\hat{x}_i\}$, so this holds iff φ_i is holomorphic. Hence $(R :_Q J) = \tilde{R}$. For $k = 1$ both sides are R .

The projection formula for the finite map η gives $H^0(S, K_S \otimes A) = H^0(\tilde{S}, \eta^*(N^* \otimes A))$. Proposition 5.1 identifies this group with $H^0(S, \Omega_{X|S}^1 \otimes A)$ when $H^0(S, \tilde{\Omega}_S^1 \otimes A) = 0$.

(c) On a reduced fibre $N^* \cong O_S$.

(d) In the test family, η^*A is trivial on the connected surface $\mathbb{P}^1 \times \mathbb{P}^1$. Also $H^0(S, \widetilde{\Omega}_S^1 \otimes A) \hookrightarrow H^0(\mathbb{P}^1 \times \mathbb{P}^1, \Omega^1) = 0$ for every λ , by the proof of Theorem 5.6. For $\lambda = 1$ the bundle A is trivial, and dt is a nonzero section. $\square$

For the candidate's cusp fibre W , whose normalization is a degree-six del Pezzo surface, the record states two things for every L :

- $\eta^*A \cong \mathcal{O}$ (the manuscript's Theorem 10.5, Step 1);
- $H^0(W, \widetilde{\Omega}_W^1 \otimes A) = 0$ (the manuscript's remark on [5] after its main theorem).

If both hold, Proposition 5.5(b) gives $H^0(W, \Omega_{X|W}^1 \otimes A) \cong \mathbb{C}$ for every L : the failure there would have size exactly one.

Status. Lemma 5.2, Propositions 5.3 and 5.5, and Corollary 5.4: proved here. Proposition 5.3 is classical in general form. Proposition 5.5 was found in the referee pass of this reader. The statement about W is conditional on the record.

5.3 The test family, audited and sharpened

Let Δ be the disc $|t| < 4/27$, let

$$C = \{([x : y : z], t) \in \mathbb{P}^2 \times \Delta : y^2z = x^2(x + z) + tz^3\},$$

put $X = C \times \mathbb{P}^1$ with $g = t$, and let $S = C_0 \times \mathbb{P}^1$ be the central fibre. Here C_0 is the nodal cubic, with node $n = [0 : 0 : 1]$. The double curve of S is $D = \{n\} \times \mathbb{P}^1$, and the normalization is $\eta = \nu \times \text{id} : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow S$. For $\lambda \in \mathbb{C}^*$, let A_λ be the line bundle on C_0 obtained by gluing the fibres of the trivial bundle over the two preimages $p = [1 : 1]$ and $q = [-1 : 1]$ of the node by λ , and put $A = \text{pr}_1^*A_\lambda$.

Theorem 5.6 (the note's Theorem 4.4). *If λ is not a root of unity, then $H^0(S, N^* \otimes A) = 0$ and $H^0(S, \widetilde{\Omega}_S^1 \otimes A) = 0$, but $H^0(S, \Omega_{X|S}^1 \otimes A) \neq 0$.*

Every step of the note's proof was checked. The recomputed parts are:

- *The total space and the fibres.* C is smooth: on $z \neq 0$, $\partial F / \partial t = -z^3$; on $z = 0$, $\partial F / \partial z = y^2 \neq 0$. The discriminant of $x^3 + x^2 + t$ is $-t(27t + 4)$, so the only singular fibre over Δ is $t = 0$.
- *The node and its normalization.* $\nu[u : v] = [(u^2 - v^2)v : u(u^2 - v^2) : v^3]$ parametrizes C_0 and sends p and q to the node. At the node, $t = UV$ with $U = y - x\sqrt{1+x}$ and $V = y + x\sqrt{1+x}$, whose Jacobian determinant at the origin is -2 .
- *The gluing bundle.* $H^0(C_0, A_\lambda) = 0$ for $\lambda \neq 1$. With the sections $O(t) = [0 : 1 : 0]$ and $P(t) = [3 : \sqrt{36+t} : 1]$, the bundle $\mathcal{O}_C(P - O)$ restricts to $A_{1/3}$ (or A_3 , by convention): the multiplier is $f(1)/f(-1) = 1/3$ for $f(r) = r - 2$, since $P(0)$ has $r = y/x = 2$. So a non-torsion A that extends to X exists (the note's Lemma 4.3).
- *The torsion at a crossing.* Take $R = \mathbb{C}\{u, v, w\}/(uv)$ and $\tau = vdu = -udv$. The kernel of $\Omega_R^1 \rightarrow$ (the differentials of the two branches) is $R\tau$, with $\tau \neq 0$ and $(u+v)\tau = 0$. Hence $\widetilde{\Omega}_S^1 \hookrightarrow \eta_*\Omega_{\mathbb{P}^1 \times \mathbb{P}^1}^1$, and $H^0(\widetilde{\Omega}_S^1 \otimes A) \hookrightarrow H^0(\mathbb{P}^1 \times \mathbb{P}^1, \Omega^1) = 0$.
- *The conductor differential.* On each branch, the pullback of $dg = vdu + u dv$ vanishes along the preimage of D . So the conductor sequence (the note's Lemma 3.1, checked in the local model) gives a unique $s \neq 0$ in $H^0(S, \Omega_{X|S}^1 \otimes A)$ with $\eta^*s = \eta^*dg \otimes e$ and $s|_D = 0$ (Proposition 3.2).

The monograph's version of this family (its Theorem 13.6, version 1.0) states that no extension of A to the total space is claimed. The note, added in version 1.1, supplies one; the multiplier $1/3$ is reproduced above.

Proposition 5.7 (the exact size of the failure). *In the test family, let $\lambda \neq 1$. Then:*

- (a) $T_S \cong O_D$, generated by $\tau = v du$;
- (b) $H^0(S, A) = H^1(S, A) = 0$;
- (c) $\delta_A = 0$, and $H^0(S, \Omega_{X|S}^1 \otimes A) \cong H^0(S, T_S \otimes A) \cong \mathbb{C}$, spanned by the conductor differential s ;
- (d) the same holds for the curve alone: $H^0(C_0, \Omega_{C|C_0}^1 \otimes A_\lambda) \cong \mathbb{C}$.

Proof. (a) At a point of D take coordinates (u, v, w) with w a coordinate on $\mathbb{P}^1$, so that $O_S = R = \mathbb{C}\{u, v, w\}/(uv)$. The note shows that the torsion there is $R\tau$. To find its annihilator, suppose $c v du = h(v du + u dv)$ in $R du \oplus R dv \oplus R dw$. Then $hu = 0$ and $(c - h)v = 0$. Since $\text{Ann}(u) = (v)$ and $\text{Ann}(v) = (u)$, this gives $c \in (u, v)$; conversely $u\tau = uv du = 0$ and $v\tau = -uv dv = 0$. So $R\tau \cong R/(u, v) = O_D$. The functions u, v come from the surface C and do not depend on the factor $\mathbb{P}^1$, so τ is defined on a whole neighbourhood of D in S , and $T_S = O_S\tau \cong O_D$.

(b) Tensor the conductor sequence with A :

$$0 \rightarrow A \rightarrow \eta_* \eta^* A \oplus A|_D \rightarrow \eta_*(\eta^* A|_{D_1 \cup D_2}) \rightarrow 0,$$

where $D_1 = \{p\} \times \mathbb{P}^1$ and $D_2 = \{q\} \times \mathbb{P}^1$. Here $\eta^* A \cong O$ with a frame e , and $A|_D \cong O_D$ with a frame σ . The ratios $e|_{D_i}/\eta^* \sigma$ are nowhere-vanishing holomorphic functions on $\mathbb{P}^1$, hence constants α, β , and $\alpha/\beta = \lambda^{\pm 1}$ by the construction of A_λ . Moreover $H^0(\mathbb{P}^1 \times \mathbb{P}^1, O) = H^0(\mathbb{P}^1, O) = \mathbb{C}$ and $H^1(\mathbb{P}^1 \times \mathbb{P}^1, O) = H^1(\mathbb{P}^1, O) = 0$ [8, Thm. III.5.1], with GAGA [16] for the analytic groups; for the quadric $\mathbb{P}^1 \times \mathbb{P}^1 \subset \mathbb{P}^3$ use $0 \rightarrow O_{\mathbb{P}^3}(-2) \rightarrow O_{\mathbb{P}^3} \rightarrow O_Q \rightarrow 0$. The long exact sequence therefore reads

$$0 \rightarrow H^0(S, A) \rightarrow \mathbb{C} \oplus \mathbb{C} \xrightarrow{\mu} \mathbb{C} \oplus \mathbb{C} \rightarrow H^1(S, A) \rightarrow 0, \quad \mu(c, k) = (\alpha c - k, \beta c - k).$$

Since $\det \mu = \beta - \alpha \neq 0$ for $\lambda \neq 1$, both groups vanish.

(c) Since S is a fibre, $N^* \cong O_S$. So δ_A maps into $H^1(S, A) = 0$, and $\delta_A = 0$. The two endpoint groups vanish: the first by (b), the second as in the note. Proposition 5.1(c) then gives $H^0(\Omega_{X|S}^1 \otimes A) \cong H^0(S, T_S \otimes A) = H^0(D, A|_D) = \mathbb{C}$, which contains $s \neq 0$.

(d) The same argument on C_0 gives $T_{C_0} = \mathbb{C}_n \tau$ and $H^i(C_0, A_\lambda) = 0$ for $i = 0, 1$. Hence the dimension is 1.

An independent count on the normalization agrees. A section of $E = \Omega_{C|C_0}^1 \otimes A_\lambda$ is a section $\tilde{\sigma}$ of $\nu^* E \cong \nu^* \Omega_C^1$ whose values at p and q agree up to the factor λ in $T_n^* C$. There is an exact sequence $0 \rightarrow N_\nu^* \rightarrow \nu^* \Omega_C^1 \rightarrow \Omega_{\mathbb{P}^1}^1 \rightarrow 0$, since ν is an immersion. The conormal bundle N_ν^* has degree $0 - (-2) = 2$, because $\deg \nu^* K_C = K_C \cdot C_0 = \deg \omega_{C_0} - C_0^2 = 0$ by adjunction. Since $H^0(\mathbb{P}^1, \Omega^1) = 0$, every global $\tilde{\sigma}$ is a section of N_ν^* , so it takes values in the conormal lines of the two branches. These are distinct at the node, since dU and dV are independent. The gluing condition therefore forces $\tilde{\sigma}(p) = \tilde{\sigma}(q) = 0$, and $\tilde{\sigma} \in H^0(N_\nu^*(-p - q)) \cong H^0(\mathbb{P}^1, O) = \mathbb{C}$. $\square$

Remark 5.8. So in the test family δ_A is not merely non-injective: its target is zero, and every torsion section lifts. Only $\lambda \neq 1$ is needed. The condition that λ is not a root of unity serves only to meet the non-torsion proviso of the published step. The factor $\mathbb{P}^1$ serves only to reach dimension three: the phenomenon is already present on the nodal fibre of a family of curves.

Proposition 5.5(d) gives the same dimension by a second route, and for every $\lambda \in \mathbb{C}^*$, including $\lambda = 1$.

Status. Theorem 5.6: audited. Proposition 5.7: proved here; the record states only the nonvanishing.

5.4 What survives: the derived global criterion

Theorem 5.9 (the note's Theorem 5.2; the monograph's Theorem 13.28). *Let X be a connected compact complex threefold with $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = 0$ and $a(X) = 1$, whose algebraic reduction is a holomorphic map onto a smooth compact curve. If some $L_0 \in \text{Pic}(X)$ has $H^2(X, T_X \otimes L_0) = 0$, then $H^i(X, T_X \otimes M) = 0$ for $i = 0, 2, 3$ and all M in a dense open subset of $\text{Pic}(X)$, and $\langle c_3(T_X), [X] \rangle \leq 0$. By the note's Corollary 5.3 it suffices that $R^2f_*(T_X \otimes L) = 0$ for one L .*

The proof in the note was checked step by step:

- $\text{Pic}(X) \cong H^1(X, O_X)$ by the exponential sequence.
- Sections vanishing to all orders along a fibre divisor vanish identically (Krull's intersection theorem), which gives $H^0(T_X(-mD)) = 0$ and $H^0(\Omega_X^1 \otimes K_X(-\ell D)) = 0$ for large m, ℓ .
- The jump loci are closed analytic subsets, by semicontinuity.
- In rational cohomology $c_1 = c_2 = 0$, and $[\text{ch}(T_X) \text{td}(X)]_3 = \frac{1}{2}c_1^3 - \frac{19}{24}c_1c_2 + \frac{1}{2}c_3$ (recomputed from Chern roots). Hence $\chi(T_X \otimes M) = \frac{1}{2}c_3 = -h^1 \leq 0$.

Two points were found in the audit:

- *Missed generality.* The hypotheses $a(X) = 1$ and the holomorphic algebraic reduction enter the proof only to produce a nonzero effective divisor D . The note's Lemma 5.1, on which the proof rests, already assumes only such a divisor.
 - *Integral cohomology.* When $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = 0$, the bundle $O_X(D)$ lies in $\text{Pic}(X) = H^1(X, O_X)$ automatically, so the same proof applies to every such threefold that carries a nonzero effective divisor.
 - *Betti numbers.* It suffices that $b_1 = b_2 = 0$. Then $\text{Pic}^0(X) = H^1(X, O_X)$ has finite index in $\text{Pic}(X)$, because $H^1(X; \mathbb{Z}) = 0$ and $H^2(X; \mathbb{Z})$ is finite. Replace D by a multiple rD with $O_X(rD) \in \text{Pic}^0(X)$, and work in the component $L_0 \cdot \text{Pic}^0(X)$. Rationally $c_1 = c_2 = 0$, and c_3 is the Euler number $2 - b_3$ [14, Cor. 11.12 and §14]. The conclusion becomes generic vanishing in that component, together with $b_3 \geq 2$.
 - *A consequence.* Let X be a compact complex threefold with $b_1 = b_2 = b_3 = 0$, for example one homeomorphic to S^6 , that carries a nonzero effective divisor. Then $H^2(X, T_X \otimes L) \neq 0$ for every $L \in \text{Pic}(X)$. By the note's Corollary 5.3, also $R^2f_*(T_X \otimes L) \neq 0$ for every L and every holomorphic map f onto a curve. So, granting $c_3(X) = 2$ and a divisor, both vanishing hypotheses fail for the candidate by this theorem alone. The record's conductor calculation adds where, and by how much, the relative one fails.
- *A citation.* The Riemann–Roch step is applied to X , which is not projective (a projective threefold has algebraic dimension 3). The note cites Hirzebruch's book [9], which proves the formula for projective manifolds. For compact complex manifolds in general it is a consequence of the Atiyah–Singer index theorem [2].

For the constructed threefold, which the record says has $c_3(X) = 2$, the contrapositive gives $H^2(X, T_X \otimes L_0) \neq 0$ for every L_0 . This is the use the record makes of the theorem.

Status. Audited. The generalizations are proved here; the first was found by this reader, the second and the consequence in its referee pass.

5.5 Relative differentials on the local models

The record corrects a remark of the source manuscript on the relative differentials of $f : X \rightarrow \mathbb{P}^1$ (the third REFUTED row, Section 4). The note's Proposition 6.1 computes three local models. The following statement contains them.

Proposition 5.10 (relative differentials of a function on a smooth germ). *Let $R = \mathbb{C}\{x_1, \dots, x_n\}$ with $n \geq 2$, let $0 \neq t$ lie in the maximal ideal $\mathfrak{m}$, and let $M_t = \Omega_{R/\mathbb{C}\{t\}}^1 = (\bigoplus_i R dx_i)/R dt$.*

- (a) *Let $\varphi = \gcd(\partial_1 t, \dots, \partial_n t)$; R is a unique factorization domain [10, Thm. 20.3]. Then $\text{Tors } M_t = R \cdot [dt/\varphi] \cong R/(\varphi)$.*
- (b) *M_t is free iff $dt(0) \neq 0$. Otherwise it needs n generators while its rank is $n - 1$.*
- (c) *If $t = u x_1^{m_1} \dots x_k^{m_k}$ with u a unit, $1 \leq k \leq n$ and all $m_i \geq 1$, then $\varphi = x_1^{m_1-1} \dots x_k^{m_k-1}$ up to a unit. In particular:*
 - *for $t = xy$ and $t = xyz$, M_t is torsion-free and not free at the origin;*
 - *for $t = s^m$ with $m \geq 2$, $\text{Tors } M_t = R/(s^{m-1}) ds$.*
- (d) *Let $n = 3$ and $\Omega_{R/\mathbb{C}\{t\}}^2 = \Omega_R^2/(dt \wedge \Omega_R^1)$.*
 - *For $t = xy$ its torsion is $R/(x, y) \cdot dx \wedge dy \neq 0$, supported on the double curve.*
 - *For $t = xyz$ its torsion is $R\zeta_1 + R\zeta_3 \cong R^2/\langle (x, 0), (y, -y), (0, z) \rangle$, where $\zeta_1 = z dx \wedge dy + y dx \wedge dz$ and $\zeta_3 = y dx \wedge dz + x dy \wedge dz$. It is $\text{Ext}_R^2(R/J, R)$ for the ideal $J = (yz, xz, xy)$ of the three coordinate axes, that is, the canonical module of that curve germ, and it needs two generators.*
 - *For $t = s^m$ with $m \geq 2$ its torsion is $(R/(s^{m-1}))^2$, on $ds \wedge dy_1$ and $ds \wedge dy_2$.*

Proof. (a) Put $w = \nabla t/\varphi$. Then $\varphi[w] = [dt] = 0$. Also $c[w] = 0$ iff $cw = d\nabla t = d\varphi w$ for some d , that is, iff $c \in (\varphi)$; so $R[w] \cong R/(\varphi)$. Conversely, let $f[a] = 0$ with $f \neq 0$, so $fa = h\nabla t = h\varphi w$ for some $h \in R$. Write $h\varphi/f = \alpha/\beta$ in lowest terms. Then $\beta a_i = \alpha w_i$ for all i , so β divides every w_i and therefore divides $\gcd(w) = 1$. So β is a unit, and $a \in R w$. (The same argument shows that R^n/Rv is torsion-free for every $v \in R^n$ whose entries have no common prime factor.)

(b) If some $\partial_i t(0) \neq 0$, then dt is part of a basis and $M_t \cong R^{n-1}$. If all $\partial_i t \in \mathfrak{m}$, then $M_t/\mathfrak{m}M_t \cong \mathbb{C}^n$, while $M_t \otimes \text{Frac } R$ has dimension $n - 1$.

(c) For $j \leq k$, $\partial_j t = x^{m-e_j}(x_j \partial_j u + m_j u)$, where the bracket is a unit. For $j > k$, $\partial_j t = x^m \partial_j u$. So every partial is divisible by $\prod_{i \leq k} x_i^{m_i-1}$. After dividing by it, the quotient for $j \leq k$ is a unit times $\prod_{i \leq k, i \neq j} x_i$. When $k = 1$ this quotient is a unit. When $k \geq 2$, a prime dividing all the quotients divides the one for $j = 1$, so it is an associate of some x_i with $2 \leq i \leq k$; but the quotient for $j = i$ is not divisible by x_i . So the quotients have no common prime factor.

(d) In the basis $dx \wedge dy, dx \wedge dz, dy \wedge dz$, the relations are $dt \wedge dx_i$.

- For $t = xy$ they are $(-x, 0, 0), (y, 0, 0)$ and $(0, y, x)$. So $\Omega_{R/\mathbb{C}\{t\}}^2 = R/(x, y) \oplus R^2/R(y, x)$. The second summand is torsion-free by the argument of (a), since x and y have no common prime factor, and the first is killed by x .
- For $t = xyz$, the map $\omega \mapsto dt \wedge \omega$ sends $\Omega_{R/\mathbb{C}\{t\}}^2$ onto $J dx \wedge dy \wedge dz$, which is torsion-free. So the torsion is $\ker(dt \wedge)/\text{im}(dt \wedge)$ on Ω_R^2 , once this quotient is known to be killed by J , which contains nonzerodivisors. Now $dt \wedge (a dx \wedge dy + b dx \wedge dz + c dy \wedge dz) = (xya - xzb + yzc) dx \wedge dy \wedge dz$. If $xya - xzb + yzc = 0$, then $x \mid c$; writing $c = xc'$ gives $y(a + zc') = zb$, so $z \mid a$; writing $a = za'$ gives $b = y(a' + c')$. Hence the kernel is free on $\zeta_1 = (z, y, 0)$ and $\zeta_3 = (0, y, x)$. In this basis the relations $dt \wedge dx, dt \wedge dy, dt \wedge dz$ are $-x\zeta_1, y(\zeta_1 - \zeta_3)$ and $z\zeta_3$, which gives the presentation. The same computation shows that $0 \rightarrow R^2 \xrightarrow{(\zeta_1 \ \zeta_3)} R^3 \xrightarrow{(xy, -xz, yz)} R \rightarrow R/J \rightarrow 0$ is exact. The cokernel of the transpose of the first map, $R^2/\langle (z, 0), (y, y), (0, x) \rangle$, is therefore $\text{Ext}_R^2(R/J, R)$, and $(u, v) \mapsto (v, -u)$ identifies it with the presentation above. So the torsion is killed by J . Since R/J is Cohen–Macaulay (a reduced curve germ), the module $\text{Ext}_R^2(R/J, R)$ is its canonical module [3, §3.3]. Its relations lie in $\mathfrak{m}$, so it needs two generators.

- For $t = s^m$ the relations are $ms^{m-1}ds \wedge dy_1$ and $ms^{m-1}ds \wedge dy_2$.

□

The global form for one-forms. This was found in the referee pass of this reader. Let $f : X \rightarrow B$ be a holomorphic map from a complex manifold onto a curve, whose fibres are locally $\{u x_1^{m_1} \cdots x_k^{m_k} = 0\}$. Put $Z = \sum_c (f^*c - (f^*c)_{\text{red}})$. Then

$$\text{Tors } \Omega_{X/B}^1 \cong (f^* \omega_B \otimes \mathcal{O}_X(Z))|_Z.$$

Proof.

- Locally Z is cut out by $\varphi_0 = \prod_i x_i^{m_i-1}$, which divides dt by Proposition 5.10(c).
- So $dt \otimes h \mapsto h dt$ is a holomorphic map $f^* \omega_B \otimes \mathcal{O}_X(Z) \rightarrow \Omega_X^1$.
- Followed by the projection to $\Omega_{X/B}^1$, its image is the torsion $R[dt/\varphi_0]$. Its kernel is $f^* \omega_B$, since $h dt \in \mathcal{O}_X dt$ forces h to be holomorphic.

For the record's threefold, granting its local models, the multiple fibres $3S_1$ and $4S_2$ give $Z = 2S_1 + 3S_2$.

Consequences for the third REFUTED row.

- The record's charts at the cusp fibre are $t = xy$ along its double curves and $t = xyz$ at its triple points. On these charts, Proposition 5.10(c) shows that $\Omega_{X/B}^1$ is torsion-free and not locally free, as the record says. The manuscript asserts torsion of $\Omega_{X/B}^1$ along all three singular fibres in its Remark 9.21, so the record's REFUTED label is correct for $p = 1$.
- The manuscript's Remark 1.3, the one named in the ledger, speaks of the sheaves $\Omega_{X/B}^p$. For $p = 2$, Proposition 5.10(d) shows that the same charts have torsion along the double curves of the cusp fibre, and the multiple-fibre charts have torsion too. So for $p = 2$ the torsion statement holds at all three singular fibres.
- Scope. These are modules on the smooth total space, not restrictions to a fibre (as the note's Remark 6.2 stresses). The multiple fibres enter through the record's description of their fillings as free quotients, with étale covers on which $t = s^m$; that description was not re-checked here.

Status. Proposition 5.10: proved here. It contains the note's Proposition 6.1, which is audited. Part (d) is new relative to the record; the note's Remark 6.2 leaves exterior powers open.

6 The later supplements, 5–9 September

The two later versions of the record add work that uses the construction's data as stated input. Both say that they do not by themselves establish the construction or a counterexample to [5] (guide 14_, its opening paragraph and §4; key-advances reader 26_, closing paragraph). This section maps them, reproduces their finite statements, and proves one arithmetic statement in full.

6.1 The topology supplement of 5 September (version 1.2)

The supplement consists of two papers, *The higher torus wrap* (15_, 50 pages) and *Real S^{12} and S^{24} routes* (16_, 64 pages), and a three-page reader's guide (14_). All three concern maps out of the cusp region of X :

1. *The angle-forgetting map.* A map $g : X \rightarrow S^3$ built from the cusp quotient is based null-homotopic. The proof exhibits an explicit framed null-bordism of the inverse image, a normally framed T^3 , inside one cusp chart times an interval. It does not use any identification of X with a sphere.
2. *The retained-angle map.* For $H : X \rightarrow S^4$ and the reconstruction's fixed diffeomorphism $d : S^6 \rightarrow X$, the record proves $[H \circ d] = \eta_4 \circ \eta_5 \neq 0$ in $\pi_6(S^4) \cong \mathbb{Z}/2$. The inverse image is a framed T^2

with the Lie framing. Its spin quadratic form $q(x, y) = x + xy + y$ over $\mathbb{F}_2$ has Arf invariant 1, since q takes the value 1 on three of the four vectors. This uses the global sphere identification as input.

3. *A Hopf morphism.* Smashing with the Hopf map, $x \mapsto \eta_2 \wedge x$, gives an injective homomorphism $\pi_6(S^4) \rightarrow \pi_9(S^6) \cong \mathbb{Z}/24$ sending $[H \circ d]$ to $12\nu_6$. This is the classical relation $\eta^3 = 12\nu$ [11] evaluated on the class of item 2, as the guide itself says.
4. *A peripheral morphism.* With the peripheral character $\chi : SL_2(\mathbb{Z}) \rightarrow \mathbb{Z}/12$ and the values $(\chi(C_0), \chi(C_1), \chi(C_2)) = (-1, 4, -3)$, the map $\Psi_H = 12\chi \bmod 24$ takes the values $(12\nu_6, 0, 12\nu_6)$, and its kernel is $\chi^{-1}(2\mathbb{Z}/12)$.
5. *Higher tori.* Over the cusp there is a local map of complex three-tori onto the two-tori, with elliptic kernel, which extends over the toric fillings. The papers do not construct a global higher-torus family. The direct rank-three cusp would have Euler characteristic 6, and with two Euler-zero finite fibres the total would be 6, not $\chi(S^8) = 2$.
6. *Octonionic cones.* The equal-diagonal positive cone of the octonionic Jordan algebra, compactified at one point, is the pair (B^{25}, S^{24}) . The papers relate it to the flag manifold $F_4/\text{Spin}(8)$; neither completes an analytic torus gluing.

Status. Items 1–3, 5 (the local map) and 6: stated; their proofs are in 15_ and 16_, which were read only in their front matter. Reproduced: the Arf invariant in item 2; the values and kernel in item 4; the Euler-characteristic arithmetic in item 5. The groups $\pi_6(S^4) \cong \mathbb{Z}/2$ (generated by $\eta_4\eta_5$) and $\pi_9(S^6) \cong \mathbb{Z}/24$, and $\eta^3 = 12\nu$, are classical [11].

6.2 The key-advances reader of 6 September

The five-page reader 26_ gives formulas and proof locations for five results. The proofs are in the frozen project archive (28_; its 782-page workbench and 138-page higher-rung paper), which was not downloaded here.

- S6-1 *The anticanonical ring recovers the fibration.* With $f^*(p_1) = 3S_1$ and $f^*(p_2) = 4S_2$, the reader states $\omega_X \cong O_X(-2S_2)$, $\omega_X^{-2} \cong f^*O_B(p_2)$ and $f_*O_X(kS_2) = O_B(\lfloor k/4 \rfloor p_2)$. Hence $\bigoplus_m H^0(X, \omega_X^{-m}) \cong \mathbb{C}[U, V]$ with $\deg U = 1$ and $\deg V = 2$, and the degree-two part recovers f .
- S6-2 *A nonzero deformation.* Replacing β by $\beta + c$, with $\text{Im } c$ below an explicit bound, gives a proper holomorphic submersion over a disc of values of c . Its Kodaira–Spencer class at each point is nonzero.
- S6-3 *The finite fillings.* There are explicit product covers of degrees nine and eight of the central surfaces of the two multiple fibres, and the kernel of the attachment map is an infinite cyclic central subgroup.
- S6-4 *Niemeier–Leech arithmetic.* Take the Niemeier lattice N with root system $A_5^4 D_4$ and glue index 72. A neighbour construction with the Weyl vector gives three neighbours, which are cycled by an order-three action. Their common intersection J has determinant 81 and index 9 in each, and the fixed extension has root system A_2^{12} with 72 roots. The value groups of an octonionic cubic on these lattices are computed, with inclusions of index 4 and 3.
- S6-5 *A common lattice in the triality space.* In Wilson’s octonionic Leech construction [12], the intersection of the Leech copy with the transported Niemeier lattices is $I = \{\sqrt{2}(a, a, a) : a \in D_4\}$, with Gram matrix $6G_{D_4}$, determinant 5184 and minimum 12. On it the cubic is $\tau(\sqrt{2}(a, a, a)) = 2\sqrt{2}P(a)$ with $P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2))$. It generates the group $4\sqrt{2}\mathbb{Z}$ but does not take the value $12\sqrt{2}$.

Proposition 6.1 (the arithmetic of S6-5). *Let $D_4 = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \equiv 0 \pmod{2}\}$, let a run over D_4 , viewed as quaternions $a_0 + a_1i + a_2j + a_3k$, and put $\tau(q) = \operatorname{Re}((qq)q)$.*

- (a) $\tau(\sqrt{2}a) = 2\sqrt{2}P(a)$.
- (b) $P(a)$ is even for every $a \in D_4$, and $P(1, 1, 0, 0) = -2$. So the values $\tau(\sqrt{2}a)$ generate $4\sqrt{2}\mathbb{Z}$.
- (c) For every $a \in \mathbb{Z}^4$, if $3 \mid P(a)$ then $9 \mid P(a)$. So P never takes a value $3k$ with $3 \nmid k$. In particular, since $\tau(\sqrt{2}a) = 12\sqrt{2}k$ means $P(a) = 6k$, $\tau(\sqrt{2}a) \neq 12\sqrt{2}k$ whenever $3 \nmid k$; the key-advances reader states the case $k = 1$.
- (d) The set of values is exactly $P(D_4) = \{n \in 2\mathbb{Z} : 3 \nmid n \text{ or } 9 \mid n\}$. So $\tau(\sqrt{2}a) = 12\sqrt{2}k$ is attained iff $3 \mid k$; for example $\tau = 36\sqrt{2}$ at $a = (-3, 0, 1, 2)$.

Proof. (a) Write $a = a_0 + v$ with v purely imaginary, so $v^2 = -|v|^2$. Then $a^2 = a_0^2 - |v|^2 + 2a_0v$, and

$$\operatorname{Re}(a^3) = a_0(a_0^2 - |v|^2) + 2a_0 \operatorname{Re}(v^2) = a_0(a_0^2 - 3|v|^2).$$

Multiply by $(\sqrt{2})^3 = 2\sqrt{2}$. The quaternions form a subalgebra of the octonions, so the octonionic product agrees.

(b) If a_0 is even, so is $P(a)$. If a_0 is odd, then $a_1 + a_2 + a_3$ is odd, so $|v|^2 \equiv a_1 + a_2 + a_3 \equiv 1$ and $a_0^2 - 3|v|^2 \equiv 1 - 3 \equiv 0 \pmod{2}$.

(c) $P(a) \equiv a_0^3 \pmod{3}$. So $3 \mid P(a)$ forces $a_0 = 3b$, and then $P(a) = 9b(3b^2 - |v|^2)$.

(d) By (b) and (c), every value lies in the set. Conversely, write $N = a_1^2 + a_2^2 + a_3^2$; the condition $a \in D_4$ is $a_0 \equiv N \pmod{2}$. Take an even n in the set.

- If $3 \nmid n$, put $k = n/2$, $a_0 = -k$ and $N = (k^2 + 2)/3$. This N is an integer since $3 \nmid k$, and then $P = -k(k^2 - k^2 - 2) = n$.
 - If k is odd, then $k^2 \equiv 1 \pmod{24}$, so $N \equiv 1 \pmod{8}$.
 - If k is even, then $N \equiv 2 \pmod{4}$.
- If $n = 18r$, put $a_0 = -3r$ and $N = 3r^2 + 2$; then $P = -3r(9r^2 - 9r^2 - 6) = n$.
 - If r is odd, then $N \equiv 5 \pmod{8}$.
 - If r is even, then $N \equiv 2 \pmod{4}$.

In each case $N \equiv a_0 \pmod{2}$, and N is not of the form $4^j(8m + 7)$, so it is a sum of three squares by the theorem of Legendre and Gauss [17, Ch. IV, Appendix]. For $n = 18$ one gets $a_0 = -3$ and $N = 5 = 0^2 + 1^2 + 2^2$. $\square$

Status. S6-1 to S6-4: stated; the proofs are in the archive 28_, which was not downloaded. Reproduced:

- S6-1: the dimension count $h^0(\omega_X^{-m}) = h^0(\mathbb{P}^1, \mathcal{O}(\lfloor m/2 \rfloor)) = \lfloor m/2 \rfloor + 1$, which matches the graded pieces of $\mathbb{C}[U, V]$. Also $\omega_X \cong \mathcal{O}_X(-2S_2)$ agrees with the manuscript's $K_X \cong f^*\mathcal{O}_B(-1) \otimes \mathcal{O}_X(2S_2)$, because $f^*\mathcal{O}_B(1) = \mathcal{O}_X(4S_2)$.
- S6-2: $\det_{\mathbb{R}} \Pi = \operatorname{Im} \tau \cdot D$, so the determinant ratio $(D + \operatorname{Im} \delta)/D$ holds.
- S6-4: the index $72 = \sqrt{6^4 \cdot 4}$; the 72 roots of A_2^{12} ; the determinant $81 = 9^2$.
- S6-5: the Gram determinant, the minimum, and the formula for τ .

Proposition 6.1: proved here; part (d) was found in the referee pass of this reader. Parts (b) and (c) are also in the reader of four workbenches [21, Prop. 4.7], which took the formula for τ from the source; part (a) checks that formula. The mod-3 step of (c) is the key-advances reader's own argument.

7 Where the record meets the owner's other workbenches

The audit note 47_ records every crossing between the owner's workbenches that the audit found, as typed edges:

- P : a proof-level import;
- I : the same identity or object on both sides;
- A : an analogy or a shared number only.

The table lists the edges that touch the S^6 material, and adds one found for this reader (marked “new”).

Edge	What crosses	Type	Status
$S^6 \rightarrow \text{YM}$	The imaginary part of the period matrix, used as a magnetic background in the Yang–Mills workbench. Only $D_{\text{YM}} = L_T q_T + 6m_T^2 = -\det_{\mathbb{R}} \Pi$ enters (Section 3)	P	audited (audit note 28_, Lemma 28.2; [20], §§6.3 and 8)
$S^6 \rightarrow \text{ES}$	The integral monodromies A_1, A_2, A_∞ of the $(3, 4, \infty)$ period system, with $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$, in the Erdős–Straus package ES-09. Nothing else from the construction is imported	P	audited for odd levels, which the application uses; false at even levels ([19], §5.2)
$\text{ES} \leftrightarrow S^6$	The Niemeier lattice with root system $A_5^4 D_4$ and glue index 72: the Erdős–Straus result R10, and S6-4 above	I	the Erdős–Straus glue code verified; the S^6 code is in the record's archive 28_, and the two were not compared
S^6 and the Jacobian map $\rightarrow \text{NS}$	Fluid coordinates, and gauge-scalar equations from the S^6 material carried to Navier–Stokes on $\mathbb{T}^3$, inside the zeta repository	P (displayed calculations)	located (audit note 47_, edge 13)
$S^6 \leftrightarrow \text{ES}$ (new)	The number $5184 = 72^2$. It is $\det I$ in S6-5, where I has Gram matrix $6G_{D_4}$, so $\det I = 6^4 \det D_4$; and it is $ \text{disc}(A_5^4 D_4) = (\det A_5)^4 \det D_4$ in S6-4 and R10	A	both products $6^4 \cdot 4$ checked; no map between the two lattices was found

Three further points of contact are not mathematical edges:

- *Hosting*. The Yang–Mills repository has an s6/ topic entry, which holds the key-advances reader, and the record's frozen edition of 9 September was released together with companion Yang–Mills and Navier–Stokes editions (guide 29_).
- *The name “Fable”*. The name labels three different things across the repositories: the three-dimensional Keller map of the Jacobian counterexample, the S^6 construction (“with Fable” in older S^6 files), and a conductor cover of the Erdős–Straus crosswalk ([19], §7). The Yang–Mills repository's attribution file, which its author designates as the clarification, says that the S^6 construction is a different object from the Jacobian counterexample, and that it was circulated by Levent Alpöge and produced with Claude.
- *The correction note and the zeta programme*. No crossing was found. The derived criterion (Theorem 5.9) and the local algebra of Section 5 use nothing from the other workbenches.

8 What remains open, and what was not checked

8.1 Open questions

1. *The construction and the counterexample.* Whether the constructed threefold exists with the stated properties, and so contradicts Theorem 2.2 of [5], needs specialist verification. This is the record's open row S6-OPEN-EXTERNAL-ADJUDICATION; its other open row asks for independent replication of the lattice, divisor and residue computations (Section 4). Engel's exposition [6] is a further independent source, and its comparison with the record has not been made.
2. *The published step.* The source manuscript (its §10.1) quotes the published sentence verbatim: from the conormal sequence, provided $L|_S$ is non-torsion, it suffices to prove the vanishing of $H^0(S, \widetilde{\Omega}_S^1 \otimes L|_S)$. Read so, the step is the implication of Section 5 with $A = L|_S$ non-torsion. The note's Lemma 4.3 shows that the test family's A can be chosen of this form on the local total space. The paper itself was not re-read here. By Corollary 5.4, the implication can fail only on non-normal fibres.
3. *The global question left by the note.* The note leaves open whether the global hypotheses of [5] force δ_A to be injective on non-normal fibres. The record's answer at its own candidate is negative: at the cusp fibre the group $H^0(W, \Omega_{X|W}^1 \otimes A)$ is nonzero for every L (the manuscript's Lemma 10.3 and Theorem 10.5; the monograph's Theorems 13.5 and 13.25). By Proposition 5.5, at a reduced fibre with normal crossings the question is exactly whether $H^0(\tilde{S}, \eta^* A) = 0$. In the test family $\delta_A = 0$ because $H^1(S, A) = 0$ (Proposition 5.7).
4. *Relative two-forms at the cusp.* The torsion of $\Omega_{X/B}^2$ on the record's charts is computed in Proposition 5.10(d): along the double curves it is the structure sheaf of the curve, and at a triple point the canonical module of the three branches. Its global form along the cusp fibre, and its effect on the direct images $R^q f_* \Omega_X^p$ through which the manuscript's Remark 1.3 would compute the Hodge numbers, were not examined. The record computes $h^{1,2} = 1$ by another route (its ledger row 17).
5. *The deformation of S6-2.* The record does not claim that different values of the parameter give non-isomorphic complex manifolds, or that the family exhausts the deformations (key-advances reader, item 2).
6. *A threefold without the candidate.* Is there a compact complex threefold with the following properties?
 - $H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = 0$, $a(X) = 1$, a holomorphic algebraic reduction, and $b_3 \geq 2$;
 - a reduced fibre with normal crossings whose normalization has a component with $H^1(O) = 0$.

By Proposition 5.5 and the argument of the manuscript's Theorem 10.5, such an X would fail the relative hypothesis of [5] for every L , independently of the S^6 construction. This question was formulated in the referee pass.

7. *Hodge numbers through relative forms.* Does the route of the manuscript's Remark 1.3, through $R^q f_* \Omega_{X/B}^p$ and the torsion computed in Proposition 5.10, reproduce $h^{1,2} = 1$? (Also formulated in the referee pass.)
8. *The two glue codes.* The Erdős–Straus workbench's order-72 glue code for $A_5^4 D_4$ and the one used in S6-4 have not been compared (Section 7).

8.2 What this reader did not check

- *The construction.* Sections 6–7 of the monograph (the period-family descent, the fillings, the gluing, the nearby-cycle and Leray comparisons) were not read. The topology and sphere

recognition, the algebraic dimension, and the Hodge, Bott–Chern and Aeppli computations were not read either. Every statement about them in this reader is the record’s.

- *The monograph beyond the parts named.* Of its 154 pages, Part I, the ledger, Theorem 13.6 and §§13.11 and 14 were read. The rest was mapped from its table of contents.
- *The later papers and the archive.* The two papers of the topology supplement (15_, 16_) were read only through their guide and their front matter. The proofs of S6-1 to S6-4 are in the archive 28_, which was not downloaded.
- *The conversation and workflow files* 07_–13_ were not used.
- *The literature.* The paper [5] was not re-read; its statements are cited from the record and the note. No novelty search was made for the additions of Section 5. The normal case of Corollary 5.4 is the source manuscript’s Remark 10.4, and the general facts used are cited where they are used.

A Catalogue of statements and files

A.1 Statements

Every statement of the record that this reader states, proves, reproduces or locates, with where it is in the record and how far it was checked. The labels are those defined in “About this reader”. File numbers refer to the record; “M” is the monograph 01_ (“item” is an item of its status list), “S” the reconstructed source manuscript 03_, and “N” the correction note 00_. Identifiers in typewriter type are rows of the claim ledger in 04_.

Statement	Where in the record	Status here	Here
<i>The construction and the counterexample claim (record: VERIFIED)</i>			
$X \rightarrow \mathbb{P}^1$ compact complex, $X \cong_{\text{diff}} S^6$, $a(X) = 1$, $c_3(X) = 2$	S, summary; M §§6–10	Located	§§3, 4
$\pi_1(X) \cong \mathbb{Z}/ 12\ell_0 - 4\ell_1 - 3\ell_2 $; the record’s triple $(0, 1, -1)$ gives -1	S §7	Located; the arithmetic reproduced	§3
Counterexample to Theorem 2.2 of [5]	M §13.11, item 5	Located	§4
Hodge diamond, Frölicher sequence; Bott–Chern and Aeppli numbers; global representatives	M, status list items 17, 19, 20	Located	§4
<i>The four rows labelled REFUTED</i>			
The singular-fibre inference of [5], p. 692 (S6-CDP-P692)	M, item 7; §13.11, item 1; N	Test family audited; see below	§5
The direct-product formula of [4], Lemma 3.2 (S6-CDP98-MONO)	M, item 12; §13.11, item 2	Located	§4
Torsion of $\Omega_{X/B}^1$ at all three singular fibres (S, Remarks 1.3 and 9.21; S6-HODGE-CUSP-TORSION)	M, item 18; §14, item 6	Proved here: correct for $p = 1$; for $p = 2$ the torsion is present	Prop. 5.10
Remark 7.28 for a fixed triple: $(100, 1, -1)$ gives 1199 (S6-LOCAL-728)	M §14, item 2	Reproduced	§4
<i>The correction note</i>			
T_S is the torsion of Ω_S^1	N, Lemma 1.1	Audited	§5
Filtration and exact criterion	N, Lemma 2.1, Props. 2.2–2.3	Audited; valid in any dimension and for vector bundles	Prop. 5.1
Normalization–conductor sequence and conductor differential	N, Lemma 3.1, Prop. 3.2, Cor. 3.3; S, Lemma 10.3	Audited	§5

Statement	Where in the record	Status here	Here
The nodal-cubic test family	N, Prop. 4.1, Lemmas 4.2–4.3, Thm. 4.4; M, Thm. 13.6	Audited; multiplier 1/3 reproduced	Thm. 5.6
Derived global criterion; relative form	N, Thm. 5.2, Cor. 5.3; M, Thm. 13.28	Audited; generalized to $b_1 = b_2 = 0$ with any nonzero effective divisor	Thm. 5.9
Local relative differentials ($t = xy, xyz, s^m$)	N, Prop. 6.1	Audited; general form proved here	Prop. 5.10
<i>New in this reader</i>			
Torsion of a hypersurface's differentials: $T_{S,x} \cong \text{Ext}^1(R/J, R)$, zero iff S is normal at x ; its global form	—	Proved here; classical in general form (Vetter, Lebelt)	Prop. 5.3
The step is valid on normal fibres	S, Remark 10.4	Located in the record; the converse proved here	Cor. 5.4
The kernel sheaf $K_S \cong \text{Hom}(\mathcal{I}, \mathcal{O}_S) \otimes N^*$; at a normal-crossing fibre the ambient group is $H^0(\tilde{S}, \eta^*(N^* \otimes A))$	—	Proved here (found in the referee pass)	Prop. 5.5
Test family: $T_S \cong \mathcal{O}_D$, $H^1(S, A) = 0$, $\delta_A = 0$, ambient group $\cong \mathbb{C}$	—	Proved here	Prop. 5.7
Torsion of $\Omega_{R/\mathbb{C}\{t\}}^1$ is $R/(\gcd \partial_i t)$, globally $(f^* \omega_B \otimes \mathcal{O}(Z)) _Z$; the torsion of relative two-forms at $t = xy, xyz$ (the canonical module of the three axes), s^m	—	Proved here (the global form found in the referee pass)	Prop. 5.10
The arithmetic of S6-5, the formula for τ on I , and the exact value set	26, item 5	Proved here; (b)–(c) also in [21, Prop. 4.7]; (d) found in the referee pass	Prop. 6.1
<i>The topology supplement (5 September)</i>			
The angle-forgetting map is null-homotopic	14; 15	Stated	§6
$[H \circ d] = \eta_4 \eta_5 \neq 0$ via Arf invariant 1	14; 15, Thm. 6.8, Cor. 6.12	Arf invariant reproduced; the rest stated	§6
Hopf morphism, $12\nu_6$; peripheral morphism Ψ_H	15, Thms. 6.13, 6.15; 16, Thms. 11.1, 12.1, 12.3	Values and kernel reproduced; classical input cited	§6
Local higher-torus map; Euler characteristic $6 \neq 2$	15	Stated; arithmetic reproduced	§6
The octonionic cone (B^{25}, S^{24})	15; 16	Stated	§6
<i>The key advances (6 September)</i>			
S6-1: the anticanonical ring $\mathbb{C}[U, V]$	26, item 1; archive	Stated; dimensions reproduced	§6
S6-2: a nonzero deformation	26, item 2; archive	Stated; determinant ratio reproduced	§6
S6-3: covers of the finite fillings; attachment kernel $\mathbb{Z}$	26, item 3; archive	Stated	§6
S6-4: Niemeier–Leech arithmetic	26, item 4; archive	Stated; 72, 72 roots, 81 reproduced	§6
S6-5: the common lattice I	26, item 5; archive	Gram data reproduced; intersection stated	§6

A.2 Files

File	Content	Use here
00_ (PDF, source zip)	The correction note	read in full; audited
01_	Verification monograph, 154 pp.	Part I, ledger, Thm. 13.6, §§13.11 and 14 read
02_	Annotated manuscript, 128 pp.	not read
03_	Clean reconstruction of the source manuscript, 108 pp.	summary, Remarks 1.3 and 9.21, §10.2 read
04_	Sources and evidence, 20.8 MB zip	its claim ledger read
05_, 06_	Mathematical provenance and its receipt	05_ read
07_–13_	Sanitised conversation and typed workflow annex, receipts, validation	not used
14_	Reader’s guide to the topology supplement	read in full
15_, 16_	The higher torus wrap (50 pp.); Real S^{12} and S^{24} routes (64 pp.)	front matter read
17_–19_	Their sources, notes and manifest	18_ read in part
26_, 27_	Key-advances reader and its source	26_ read in full
28_	Frozen project, 174 MB, 1,079 files	not downloaded
29_, 30_	Guide and manifest of the frozen project	29_ read
README.md; metadata, rights, licence and checksum files	Version 1.0 front file; record metadata	README.md read

B How this reader was checked

Sources. The record 10.5281/zenodo.22678442. Its file list and MD5 checksums, and the files added by each of its four versions, were read from the record’s public metadata on 26 September 2026. Thirty-two of its 33 files were downloaded, with the owner’s permission, and their MD5 checksums and sizes equal the record’s; the 174 MB archive 28_ was not downloaded. The script `s6r_record_inventory.py` reproduces this. The parts of the record read, and those only mapped, are listed in Appendix A.

What was refereed earlier. The overlap edges of Section 7 other than the one marked “new” come from the audit notes 28_ and 47_. These were refereed (the ninth referee pass, and the referee pass on the notes 43_–47_) and entered the reader of four workbenches [21].

The referee pass of this reader. On 27 September 2026 an independent Claude instance that had written none of the text refereed the version of 26 September. It read in both directions: for overstatement, and for understatement, missed generality and missed or implied results. It reran both scripts of this reader and wrote its own checks, 22 of them, all passing; they are included as `referee/referee_checks.py`. It found no mathematical error. It made 2 major and 12 minor findings.

- *Major findings.*

- The abstract listed as new that the step is valid on normal fibres, which is the source manuscript’s Remark 10.4.
- The record’s two OPEN rows were misreported.

- *Minor findings.* Among them:

- the classical results of Vetter and Lebelt were not cited;
- the phrase “on any fibre” should read “on any reduced fibre”;
- the example gluing triple was the reader’s own, not the record’s;
- the description of the monograph’s §14 was inaccurate;
- Engel’s citation is of the 1998 paper;
- S6-1 to S6-4 were labelled “located” although their proofs were not seen.

Every finding was checked by the author against the sources before it was applied. For example, the claim ledger’s two open rows were read in the evidence archive, and Engel’s text and the bibliographic data of Vetter, Lebelt and Miller–Vassiliadou were read online. All findings were applied.

The pass also proposed missed or implied results, each with a proof. The author verified each proof, and each result is in this version with its credit:

- Proposition 5.5;
- the global form of the torsion of relative one-forms;
- the extension of Theorem 5.9 to $b_1 = b_2 = 0$, with its consequence for rational homology six-spheres;
- the exact value set in Proposition 6.1(d), checked numerically for $|n| \leq 3000$;
- two open questions (Section 8).

Two further proposals were not adopted. A reflexivity criterion was left out as outside the reader’s topic. A sketched global form of the torsion of relative two-forms was left out because only a sketch was given.

What rests on this author and one referee pass.

- the map of the record (Sections 2, 4 and 6);
- the audit of the correction note (Section 5);
- Lemma 5.2, Propositions 5.3, 5.5, 5.7, 5.10 and 6.1, and Corollary 5.4;
- the generalizations of Theorem 5.9 and the remark on its citation;
- the new overlap edge.

The scripts of this reader. They import no code from the record. They are in `checks/s6_reader/` on the branch named below.

Script	What it checks	Time, result
s6r_checks.py	47 symbolic and exact checks, in four groups: • <i>the test family</i>: smoothness of the total space, the discriminant, the normalization and the local model at the node, the sections O, P and the multiplier $1/3$, $H^0(A_\lambda) = 0$, the relations $u\tau = v\tau = 0$, the map μ of Proposition 5.7, and the independence of dU, dV; • <i>the local modules</i>: the gcd formula on nine monomials with unit factors, the torsion of relative one- and two-forms at $t = xy, xyz, s^m$, and the presentation of the torsion at $t = xyz$; • <i>the derived criterion</i>: the Riemann–Roch coefficient $\frac{1}{2}$ of c_3; • <i>the finite statements</i>: 1199; the Arf invariant; Ψ_H; the Euler characteristic; $\det_{\mathbb{R}} \Pi = \operatorname{Im} \tau \cdot D$; the numbers of S6-1, S6-2, S6-4 and S6-5, including the quaternion formula for τ.	2 s; all pass
s6r_record_inventory.py	the 33 files of the record against the downloaded copies, by MD5 and size	under 1 s; 32 match, one not local
referee/referee_checks.py	the referee’s own checks: the kernel sheaf at a triple point, Hilbert functions of the relative forms at $t = xy, xyz, x^2y$ and two quadrics, the value set of S6-5, the source’s matrices, the gluing triples and the claim-ledger counts	7 s; 22 checks, all pass

What has not been done. The new results of this version, Proposition 5.5 among them, have been checked by the author and by the one referee pass that proposed or read them, but not by a second independent pass. The paper [5] itself, the record’s construction, and the archive 28_ were not read (Section 8).

Where everything is. This reader, its sources and its scripts are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, in `contrib/claude-ab/grind_20260925/`: the reader in `s6_reader/`, the scripts in `checks/s6_reader/` (the referee’s in `checks/s6_reader/referee/`, with its report), and the audit notes at the top level.

References

- [1] *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, manuscript circulated by L. Alpöge, 2026, <https://alpo.ge/s6.pdf>. Its opening summary is titled *The $(3, 4, \infty)$ modular family of 2-tori, completed at its three special points, is a complex structure on S^6* .
- [2] M. F. Atiyah, I. M. Singer, *The index of elliptic operators: III*, Ann. of Math. (2) **87** (1968), 546–604.
- [3] W. Bruns, J. Herzog, *Cohen–Macaulay Rings*, Cambridge Studies in Advanced Mathematics **39**, Cambridge University Press, Cambridge, 1993.
- [4] F. Campana, J.-P. Demailly, T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compositio Math. **112** (1998), 77–91.
- [5] F. Campana, J.-P. Demailly, T. Peternell, *The algebraic dimension of compact complex threefolds with vanishing second Betti number*, Compositio Math. **156** (2020), 679–696; arXiv:1904.11179.
- [6] P. Engel, *Complex structures on S^6* , 13 September 2026, <https://philip-engel.github.io/S6.pdf>.
- [7] H. Grauert, R. Remmert, *Coherent Analytic Sheaves*, Grundlehren der mathematischen Wissenschaften **265**, Springer, Berlin, 1984.

- [8] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics **52**, Springer, New York, 1977.
- [9] F. Hirzebruch, *Topological Methods in Algebraic Geometry*, 3rd ed., Grundlehren der mathematischen Wissenschaften **131**, Springer, Berlin, 1966.
- [10] H. Matsumura, *Commutative Ring Theory*, Cambridge Studies in Advanced Mathematics **8**, Cambridge University Press, Cambridge, 1986.
- [11] H. Toda, *Composition Methods in Homotopy Groups of Spheres*, Annals of Mathematics Studies **49**, Princeton University Press, Princeton, 1962.
- [12] R. A. Wilson, *Octonions and the Leech lattice*, J. Algebra **322** (2009), 2186–2190.
- [13] K. Lebelt, *Torsion äußerer Potenzen von Moduln der homologischen Dimension 1*, Math. Ann. **211** (1974), 183–197.
- [14] J. W. Milnor, J. D. Stasheff, *Characteristic Classes*, Annals of Mathematics Studies **76**, Princeton University Press, Princeton, 1974.
- [15] C. Miller, S. Vassiliadou, *(Co)torsion of exterior powers of differentials over complete intersections*, J. Singul. **19** (2019), 131–162.
- [16] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Ann. Inst. Fourier **6** (1956), 1–42.
- [17] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics **7**, Springer, New York, 1973.
- [18] U. Vetter, *Äußere Potenzen von Differentialmoduln reduzierter vollständiger Durchschnitte*, Manuscripta Math. **2** (1970), 67–76.
- [19] Claude (Anthropic), *The Erdős–Straus workbench: a reader of its results, with proofs*, 2026, version before its independent referee pass; source on the branch `claude/claude-ab-grind-20260925` of KokunoYumeto/zeta-function-research-reader, folder `contrib/claude-ab/grind_20260925/es_reader/`.
- [20] Claude (Anthropic), *The Yang–Mills workbench: a reader of its results, with proofs*, 2026, version before its independent referee pass; same branch, folder `ym_reader/`.
- [21] Claude (Anthropic), *Four workbenches and where they meet: a reader of the Erdős–Straus, Collatz, Erdős Problem 817 and Yang–Mills workbenches*, 2026, file `00-claude-opus-5-5-workbench-reader.pdf` of the Zenodo record DOI 10.5281/zenodo.22977243; source on the same branch, folder `workbench_reader/`.
- [22] Claude (Anthropic), *The split-zero programme around the Riemann zeta function: a reader of its results, with proofs*, version 2, Zenodo, 2026, DOI 10.5281/zenodo.22977243.